

C2 Event Classification with Neural Networks

Jet-flavour tagging at FCC-ee

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Talk roadmap (15 min)

- 1 Physics Background
- 2 Dataset
- 3 Pre-processing
- 4 Model
- 5 Experiments and Performance
- 6 Discussion
- 7 Conclusion

From e^+e^- to the detected particles

1 e^+e^- collision

FCC-ee sets the centre-of-mass energy to $\sqrt{s} = 91.2 \text{ GeV}$, equal to the mass of the Z^0 . The produced Z^0 is therefore almost at rest in the detector.

2 Prompt Z^0 decay

The Z^0 lives for $\sim 3 \times 10^{-25} \text{ s}$ and immediately decays into a quark–antiquark pair

$$Z^0 \rightarrow q \bar{q} \quad (q = b, c, s).$$

3 Parton shower

Quarks cannot exist in isolation: as they separate, each emits gluons and sometimes produces light $q\bar{q}$ pairs, spreading its energy among coloured partons. The resulting narrow cascade forms a *jet*.

4 Hadronisation

When parton energies drop to $\mathcal{O}(1) \text{ GeV}$ the coloured partons combine into colour-neutral hadrons (mesons and baryons).

5 Decays to long-lived states

Most hadrons are unstable and decay within millimetres. Only the following particles survive long enough to reach the sub-detectors:

$$\gamma, e^\pm, \mu^\pm, \pi^\pm, K^\pm, K_L^0, p, \bar{p}, n, \bar{n}.$$

Detector concept

- **Tracking system**

Record the curved paths of charged particles, giving momentum, charge sign and precise vertex positions.

- **Electromagnetic calorimeter (ECAL)**

Stops $e^\pm$ and γ ; their electromagnetic showers reveal the particle energies.

- **Hadronic calorimeter (HCAL)**

Absorbs hadrons ($\pi^\pm$, $K^\pm$, p , $\bar{p}$, n , $\bar{n}$, ...) and measures their energy through hadronic cascades.

- **Muon spectrometer**

Outer chambers tag the penetrating $\mu^\pm$ and determine their momentum.

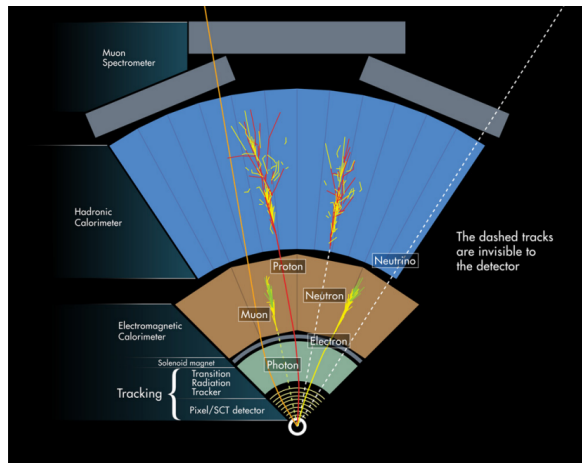


Figure 1: Sketch of the layered detector geometry.[1]

Dataset overview

- Three Monte-Carlo samples at FCC-ee Z -pole: Z_{bb} , Z_{cc} , Z_{ss} [2].
- **# events**: $\sim 200\text{k}$ per flavour $\Rightarrow 50\text{k} / 15\text{k} / 15\text{k}$ for train / val / test.
- Each event provides two tables
 - **Particle records** : E , n_{particle} , m , q , p , p_T , η , ϕ , (p_x, p_y, p_z) , ID, origin-vertex index, thrust error, angle to thrust.
 - **Vertex records**: track multiplicity, fit χ^2 , primary flag, invariant mass, (x, y, z) position and combined error.
- Files list only *final-state* particles; intermediate hadrons are gone. Their presence is instead can be inferred from the vertex-mass spectra shown next.

Vertex-mass fingerprints

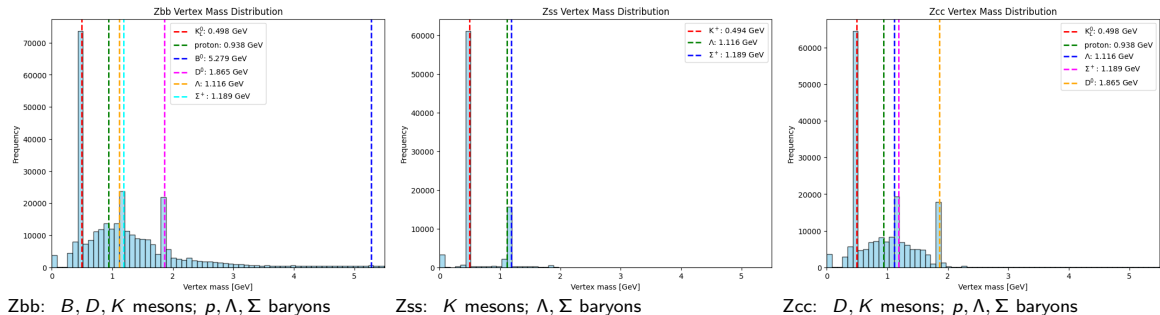


Figure 2: Peaks highlight typical hadrons from each $q\bar{q}$ flavour.

Pre-processing pipeline

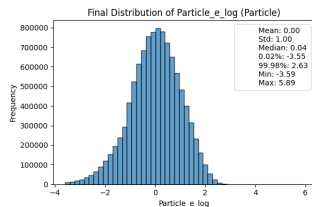
- 1 Shuffle events; fill remaining NaNs with the column median.
- 2 log-transform high-dynamic-range variables (E , p , p_T , errors, χ^2).
- 3 Encode angles ϕ , θ_T as $(\sin, \cos)$.
- 4 Clip values to the 0.02–99.98 percentile window.

- 5 Z-score features:

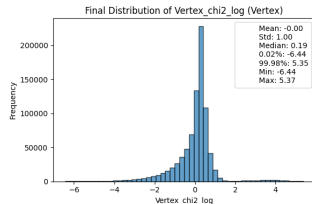
$$x \longrightarrow \frac{x - \mu}{\sigma}, \quad (\mu, \sigma) \text{ from training set.}$$

- 6 **Particles**: keep 35 charged tracks (p_T -sorted); if < 35 add top- p_T neutrals, then pad -999.
- 7 **Vertices**: keep 5 with most tracks; if < 5 pad -999.
- 8 **Two data variants**: with Particle_ID & origin vertex index vs. without them.
- 9 Resulting shapes: Particles (35, 12–14), Vertices (5, 8).

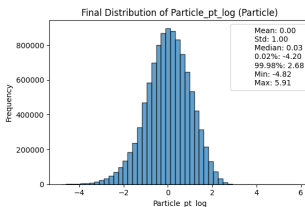
Example cleaned distributions



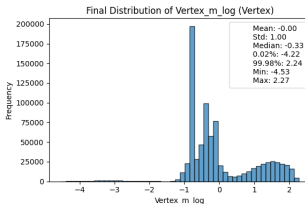
Particle $\log(E)$



Vertex $\log(\chi^2)$



Particle $\log(p_T)$



Vertex $\log(m)$

Figure 3: Representative particle- and vertex-level feature distributions after all cleaning steps.

Model overview

- **Two-stream input** particles (35, 12–14) + vertices (5, 8).
- Each feature row is linearly embedded to a shared size $d_{\text{model}} = 64$; embeddings are concatenated into a length-40 sequence.
- **Mask** skips pad entries -999 in every attention layer.
- **4 Transformer encoder layers**: 6-head self-attention + FFN ($d_{\text{ff}} = 128$) with residual and layer-norm.
- Conv1D $\times 32$ (kernel size of three) $\rightarrow$ global average pool.
- Dense 64 $\rightarrow$ softmax 3 (flavours b, c, s).
- Training: 50 epochs, categorical cross-entropy, Adam ($\text{lr} = 10^{-3}$) with gradient clip $\|g\| < 1$ and early-stopping on validation loss.
- Initial test with $d_{\text{model}} = 128$ & 6 layers diverged (exploding grads) $\rightarrow$ current depth chosen for stability.

Transformer flavour-tagger architecture

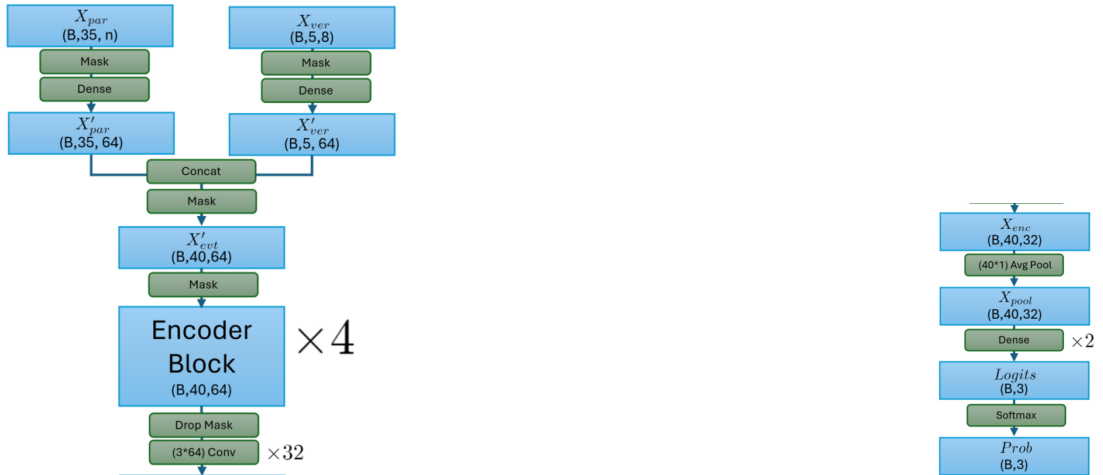


Figure 4: Two-stream Transformer tagger.

Experiments and performance

- **Two variants**

- **Exp 1 – with ID:** includes Particle_ID and origin-vertex index.
- **Exp 2 – no ID:** excludes both; relies solely on measurable quantities.
- Raw integer Particle_IDs behave like arbitrary continuous numbers; without proper one-hot or learned embedding they mis-scale the features and can *degrade* generalisation.
- **Accuracy comparison** (train / validation / test).

Experiment	Train acc	Val acc	Test acc
Exp 1 (with ID)	0.905	0.903	0.909
Exp 2 (no ID)	0.933	0.926	0.927

Table 1: Classification accuracy of the two model variants.

Training curves

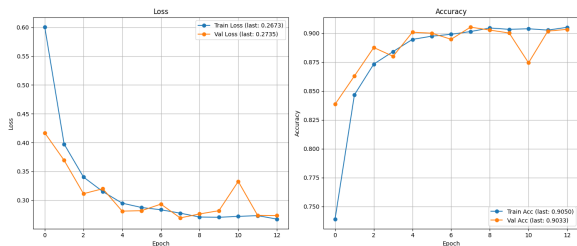


Figure 5: Exp. 1 (with ID) – early stop at epoch 12

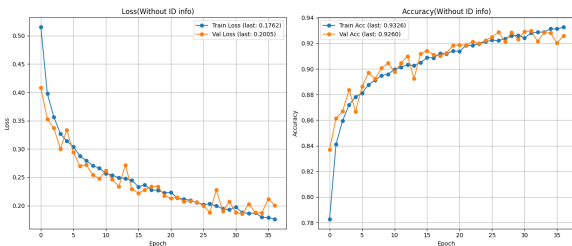


Figure 6: Exp. 2 (no ID) – early stop at epoch 36

The following slides focus on the performance of the **no-ID model**.

Train–test output: KS test

- Compare predicted–probability spectra for **train** vs. **test**.
- Kolmogorov–Smirnov p -values:
 $p_{Zbb} = 0.31$, $p_{Zss} = 0.94$,
 $p_{Zcc} = 0.97$.
- All $p > 0.05 \Rightarrow$ train and test distributions match within statistical fluctuations — no observable signs of over-fitting.

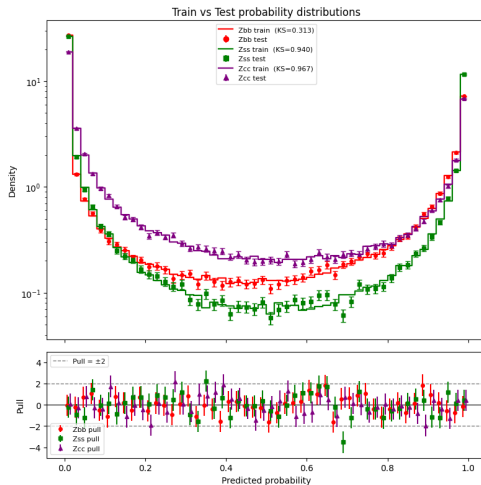


Figure 7: Train (histograms) vs. test (points) probability spectra.

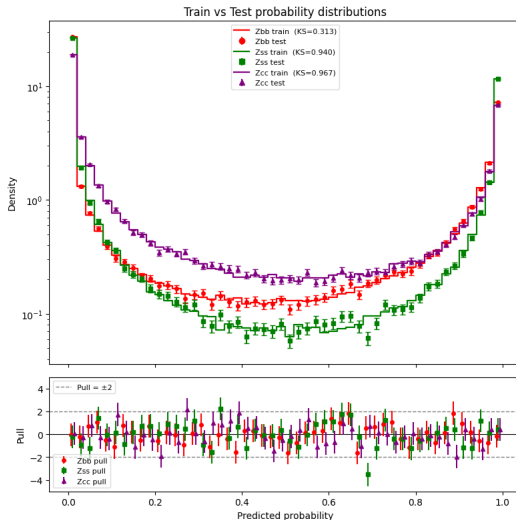
Statistical details: Poisson error and pull

- **Poisson error:** for each bin $\sigma = \sqrt{N}$ since $\text{Var}(N) = N$.

- **Pull definition**

$$\text{pull}_i = \frac{N_i^{\text{train}} - N_i^{\text{test}}}{\sqrt{N_i^{\text{train}} + N_i^{\text{test}}}}.$$

- Pull values cluster around zero; most bins have $|\text{pull}| < 2$, indicating stable train-test agreement with no observable sign of over-fitting.



Per-flavour probability densities — no-ID model

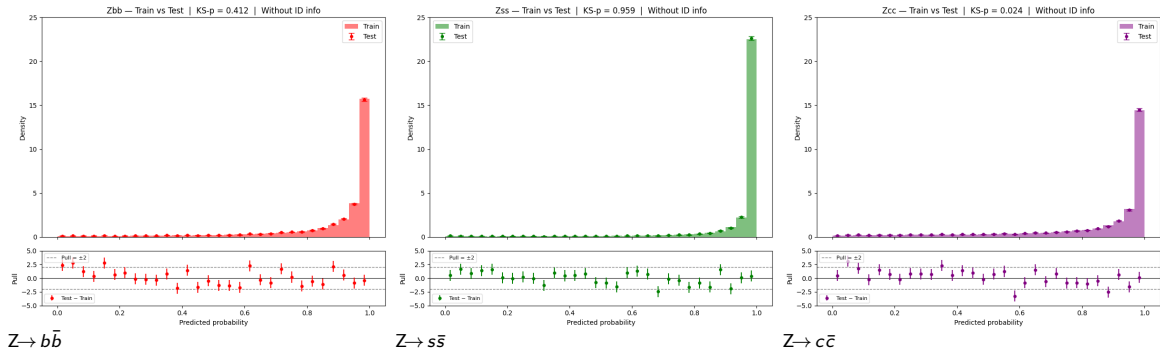


Figure 9: Top: train bars vs. test points (Poisson errors), linear scale 0–25; Bottom: per-bin pull.

Higher peak at probability 1 $\Rightarrow$ better per-jet accuracy; narrower curve $\Rightarrow$ more consistent predictions. s -jets show the tallest, narrowest peak; b -jets come next; c -jets remain the most ambiguous.

Confusion matrix — 15 k test events / flavour

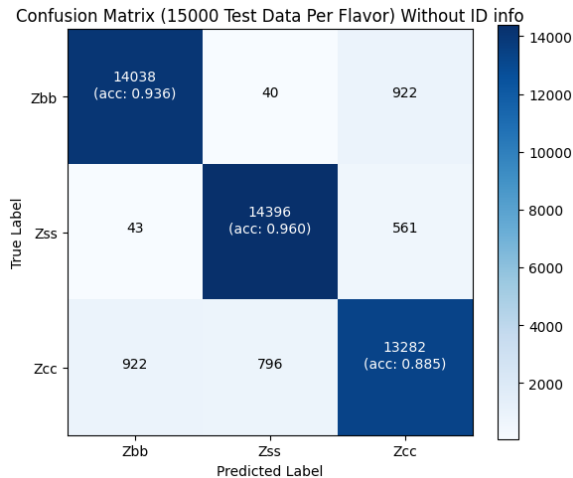


Figure 10: Confusion matrix of the no-ID model on 15 k test events per flavour. Overall test accuracy = 0.927; most mis-tags occur between *b* and *c*, while *s* is cleanly identified.

ROC curves — one-vs-rest, no-ID model

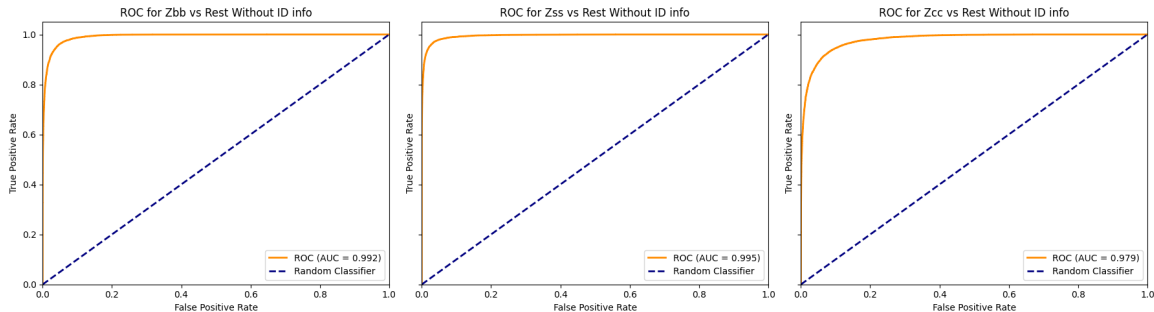


Figure 11: One-vs-rest ROC curves for the no-ID model

True positive rate (TPR) and false positive rate (FPR) are defined as:

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

AUC: *b* 0.992, *s* 0.995, *c* 0.979

- **Overall performance** No-ID model attains **92.7 %** test accuracy on this 3-class FCC dataset. (For context, current state-of-the-art taggers achieve about 86% accuracy on harder multi-flavour jet-tagging datasets.[3].)
- **Why might the FCC Z -pole be “easier”? — working hypotheses**
 - ① Z^0 produced almost at rest $\Rightarrow$ clean back-to-back jets.
 - ② Decay features (e.g. vertex mass, track multiplicity) differs for b, c, s , providing critical discrimination.
 - ③ Only three flavours considered; no other backgrounds.
- **s -jets appear easiest** Vertex-mass spectrum is very clean, dominated by a sharp K peak (0.5 GeV) and a distinct Λ/Σ bump (1.1 GeV).
- **Main confusion: b vs. c** Their vertex-mass distributions largely overlap, so most mis-tags occur here; a slight edge for b may stem from the longer B -hadron decay chain, but this remains a hypothesis.

Vertex-mass fingerprints

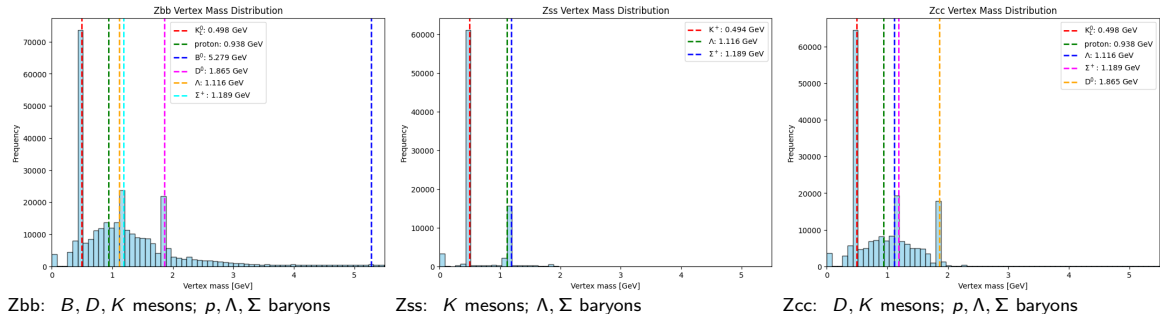


Figure 12: Peaks highlight typical hadrons from each $q\bar{q}$ flavour.

Ablation study: `vertex_mass` and n_{tracks}

- Event-level vertex mass and track count are strongly correlated ($\rho_{\text{Spearman}} = 0.77$).
- **Ablation check** Removing `vertex_mass` and its correlated n_{tracks} reduces accuracy by up to 5 pp, supporting hypothesis (2).

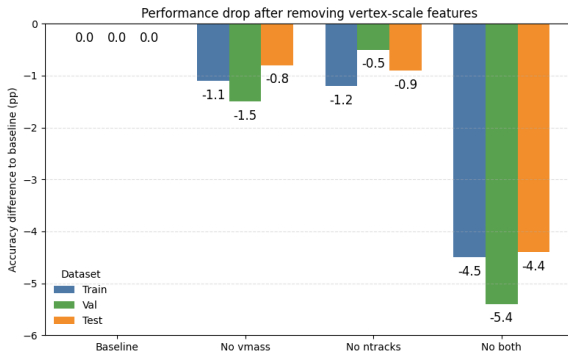


Figure 13: Removing either feature costs $\sim 1\%$ drop; deleting both cuts accuracy by about 4.5% . Vertex mass and track multiplicity jointly carry key information.

- **Dataset:** FCC-ee Z -pole simulation offers clean back-to-back jets and large statistics for heavy-flavour studies.
- **Method:** Two-stream Transformer + attention mask handles variable-length particle/vertex inputs *without* bespoke feature engineering.
- **Result:** Using only detector-level observables achieves **92.7 %** test accuracy; adding truth-level IDs degraded generalisation.
- **Interpretation:** Vertex-mass spectra and vertex track multiplicity provide *critical* flavour information— s jets are most distinct, while b and c overlap.
- **Next step:** Apply SHAP feature-attribution to quantify which variables drive the classifier decisions.

References I



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Thank you!

Any questions?